TRANSPORTATION MODELS: NORTHWEST-CORNER METHOD

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Some special items are produced at factories at Patna, Guwahati, Mumbai and Hyderabad. Distributors are located at Lucknow, Delhi, Jaipur, Chennai and Kolkata. Supply and demands of items are indicated in table. Note that factories have capacity to produce 18,000 items, but only 15,000 items are in demand. Transportation cost of each item is proportional to distance it travels. The road distances from factory to distribution centers (in km) are indicated in each cell. Your goal is to minimize the transportation cost.

Distributor Factory	Lucknow	Delhi	Jaipur	Chennai	Kolkata	Supply
Patna	610	1,060	1,180	1,490	580	3,000
Guwahati	1,020	1,910	1,946	2,700	1,018	6,000
Mumbai	1,370	1,420	1,150	1,330	2,050	4,000
Hyderabad	1,300	1,560	1,470	630	1,500	5,000
Demand	2,000	7,000	4,000	1,000	1,000	

Unbalanced Problem: add dummy destination

Distributor Factory	Lucknow	Delhi	Jaipur	Chennai	Kolkata	Dummy	Supply
Patna	610	1,060	1,180	1,490	580	0	3,000
Guwahati	1,020	1,910	1,946	2,700	1,018	0	6,000
Mumbai	1,370	1,420	1,150	1,330	2,050	0	4,000
Hyderabad	1,300	1,560	1,470	630	1,500	0	5,000
Demand	2,000	7,000	4,000	1,000	1,000	3000	

Stepwise Procedure

Step 1

	Lucknow	Delhi	Jaipur	Chennai	Kolkata	D_{dummy}	Supply
Patna	610(2000)	1060	1180	1490	580	0	1000
Guwahati	1020	1910	1946	2700	1018	0	6000
Mumbai	1370	1420	1150	1330	2050	0	4000
Hyderabad	1300	1560	1470	630	1500	0	5000
Demand	0	7000	4000	1000	1000	3000	

Step 2

	Lucknow	Delhi	Jaipur	Chennai	Kolkata	D_{dummy}	Supply
~~Patna~~	~~610(2000)~~	~~1060(1000)~~	~~1180~~	~~1490~~	~~580~~	~~0~~	0
Guwahati	1020	1910	1946	2700	1018	0	6000
Mumbai	1370	1420	1150	1330	2050	0	4000
Hyderabad	1300	1560	1470	630	1500	0	5000
Demand	0	6000	4000	1000	1000	3000	

Step 3

	Lucknow	Delhi	Jaipur	Chennai	Kolkata	D_{dummy}	Supply
~~Patna~~	~~610(2000)~~	~~1060(1000)~~	~~1180~~	~~1490~~	~~580~~	~~0~~	0
~~Guwahati~~	~~1020~~	~~1910(6000)~~	~~1946~~	~~2700~~	~~1018~~	~~0~~	0
Mumbai	1370	1420	1150	1330	2050	0	4000
Hyderabad	1300	1560	1470	630	1500	0	5000
Demand	0	0	4000	1000	1000	3000	

Step 4

	Lucknow	Delhi	Jaipur	Chennai	Kolkata	D_{dummy}	Supply
~~Patna~~	~~610~~(2000)	~~1060~~(1000)	~~1180~~	~~1490~~	~~580~~	~~0~~	0
~~Guwahati~~	~~1020~~	~~1910~~(6000)	~~1946~~	~~2700~~	~~1018~~	~~0~~	0
~~Mumbai~~	~~1370~~	~~1420~~	~~1150~~(4000)	~~1330~~	~~2050~~	~~0~~	0
Hyderabad	1300	1560	1470	630	1500	0	5000
Demand	0	0	0	1000	1000	3000	

Step 5

	Lucknow	Delhi	Jaipur	Chennai	Kolkata	D_{dummy}	Supply
~~Patna~~	~~610~~ **~~(2000)~~**	~~1060~~ **~~(1000)~~**	~~1180~~	~~1490~~	~~580~~	~~0~~	0
~~Guwahati~~	~~1020~~	~~1910~~ **~~(6000)~~**	~~1946~~	~~2700~~	~~1018~~	~~0~~	0
~~Mumbai~~	~~1370~~	~~1420~~	~~1150~~ **~~(4000)~~**	~~1330~~	~~2050~~	~~0~~	0
Hyderabad	1300	1560	1470	630	1500	0	5000
Demand	0	0	0	1000	1000	3000	

Step 6

	Lucknow	Delhi	Jaipur	Chennai	Kolkata	D_{dummy}	Supply
~~Patna~~	~~610~~(~~2000~~)	~~1060~~(~~1000~~)	~~1180~~	~~1490~~	~~580~~	~~0~~	0
~~Guwahati~~	~~1020~~	~~1910~~(~~6000~~)	~~1946~~	~~2700~~	~~1018~~	~~0~~	0
~~Mumbai~~	~~1370~~	~~1420~~	~~1150~~(~~4000~~)	~~1330~~	~~2050~~	~~0~~	0
Hyderabad	1300	1560	1470	630(1000)	1500	0	4000
Demand	0	0	0	0	1000	3000	

Step 7

	Lucknow	Delhi	Jaipur	Chennai	Kolkata	D_{dummy}	Supply
~~Patna~~	~~610~~(2000)	~~1060~~(1000)	~~1180~~	~~1490~~	~~580~~	~~0~~	0
~~Guwahati~~	~~1020~~	~~1910~~(6000)	~~1946~~	~~2700~~	~~1018~~	~~0~~	0
~~Mumbai~~	~~1370~~	~~1420~~	~~1150~~(4000)	~~1330~~	~~2050~~	~~0~~	0
Hyderabad	1300	1560	1470	630(1000)	1500(1000)	0	3000
Demand	0	0	0	0	0	3000	

Step 8

	Lucknow	Delhi	Jaipur	Chennai	Kolkata	D_{dummy}	Supply
Patna	610(2000)	1060(1000)	1180	1490	580	0	0
Guwahati	1020	1910(6000)	1946	2700	1018	0	0
Mumbai	1370	1420	1150(4000)	1330	2050	0	0
Hyderabad	1300	1560	1470	630(1000)	1500(1000)	0(3000)	0
Demand	0	0	0	0	0	0	

	Lucknow	Delhi	Jaipur	Chennai	Kolkata	D_{dummy}	Supply
Patna	610 **(2000)**	1060 **(1000)**	1180	1490	580	0	3000
Guwahati	1020	1910 **(6000)**	1946	2700	1018	0	6000
Mumbai	1370	1420	1150 **(4000)**	1330	2050	0	4000
Hyderabad	1300	1560	1470	630 **(1000)**	1500 **(1000)**	0 **(3000)**	5000
Demand	2000	7000	4000	1000	1000	3000	

The minimum total transportation cost

=610×2000+1060×1000+1910×6000+1150×4000+630×1000+1500×1000+0×3000

= 20,470,000.

Degenerate basic feasible solution

- For m sources and n destination, there are $m+n$ constraint equations.
- However, as transportation model is always balanced, number of independent constraint equations is $m+n-1$.
- It means that starting basic feasible solution should have $m+n-1$ basic variables.
- A degenerate basic feasible solution (BFS) in linear programming is one where at least one of the basic variables (variables in the basis) is equal to zero. In a non-degenerate BFS, all basic variables are strictly positive.
- Previous example shows degenerate basic feasible solution.

Resolving degeneracy

- Degeneracy may occur at (1) Initial stage or (2) at subsequent iterations.
- To resolve degeneracy, we make use of an artificial quantity (d=0).
  The quantity d is assigned to that unoccupied cell, which has the minimum transportation cost.

	Lucknow	Delhi	Jaipur	Chennai	Kolkata	D_{dummy}	Supply
Patna	610 **(2000)**	1060 **(1000)**	1180	1490	580 (0=d)	0	3000
Guwahati	1020	1910 **(6000)**	1946	2700	1018 (0=d)	0	6000
Mumbai	1370	1420	1150 **(4000)**	1330	2050	0	4000
Hyderabad	1300	1560	1470	630 **(1000)**	1500 **(1000)**	0 **(3000)**	5000
Demand	2000	7000	4000	1000	1000	3000	

Degeneracy can be resolved by following these steps:

- **Northwest-corner method:** This method starts at the northwest-corner cell of the table (variable x_{11})
- **Step 1:** Allocate as much as possible to the selected cell. Adjust the associated amounts of supply and demand by subtracting the allocated amount.
- **Step 2:** Cross out the row and column with zero remaining supply or demand. It indicates that no further assignments can be made in that row or column. If both row and a column net to zero simultaneously, cross out only one. Leave a zero supply (demand) in the uncrossed out row (column)

Degenerate basic feasible solution

- **Step 3: If exactly one row or column is left uncrossed out, stop.**

Otherwise move to the cell to the right if a column has just been crossed or the one below if a row has been crossed out. Go to step 1.

(Based on book by H.A. Taha).